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With their sleek lines, thin profile, light weight and flat screens, LCD monitors have always caught one's fancy. Their high costs limited them to niche segments, though, but due to several factors, prices of LCD screens are plummeting. The smaller ones have now come within the reach of home users, and LCDs seem poised to replace CRTs for good.

There is a variant in the LCD space which must be talked about here. Regular CRT monitors and LCD panels come in what is termed the 4:3 aspect ratio. Your television set, for example, has a 4:3 display screen. Some displays, however, come in a 16:9 screen ratio, which means the width of the screen is significantly more than its height. Such a display is called a "widescreen," and makes for a better movie-watching experience.

While the widescreen aspect is another positive on the LCD's side, viewing angles have been a problem: an LCD tends to appear darker and discoloured when viewed beyond a certain angle. This is undesirable when the screen is being viewed by number of people seated at different positions—as would be the case if they were watching a movie. Some displays today apply a special coat to their LCD panels, which goes a long way in increasing the viewing angle and the image contrast ratio (more on contrast ratio later).

Before you pick an LCD monitor, we must point out another

potential roadblock to its enjoyment. It might have a slow response time, which can lead to the appearance of trails, or what is known as "ghosting" of images. Ghosting is generally visible as trails behind fast-moving objects. It is a somewhat subjective effect—some people can see it, others cannot. It is thus a good idea to try out a movie or a game on an LCD panel before purchasing it.

We must tell you that LCD monitors aren't yet perfect for movies or games. Thankfully, at least one issue seems to be resolved: buyers were often told to beware of dead pixels before picking an LCD screen. (A dead pixel is one that is either always on or off, and is thus useless in general, everyday use.) With better quality control, however, dead pixels are rarely seen on LCD panels today.

Here, we compare a range of available LCD monitors. We received a total of 33 monitors for this test. Of these, the 15-inch category consisted of seven units from seven brands, the 17-inchers consisted of 15 from 12 brands, while there were 11 19-inch monitors from nine brands. NEC sent in the highest number of LCDs at seven. Our effort here has been not to help you make a well-informed purchase decision.

A Look At Features

Let us first take a look at some of the key features to look at before you zero in on an LCD. These are by no means the only features there are, but are certainly the most important aspects of such a display unit.

Pixel Pitch

The distance between two white pixels or two sub-pixels of the same colour is known as the pixel pitch of the monitor. A white pixel is made up of three sub-pixels—one red, one green and one blue. All the categories saw pixel pitches of 0.29mm and 0.26mm. Pixel pitch varies with monitor size, and is generally fixed across a size category. Needless to say, the individual size of each sub-pixel is smaller than the pixel pitch.

Native Resolution And Screen Aspect Ratio

LCD monitors have what is called a "native resolution," and cannot display crisp images at a resolution other than this resolution—unlike CRT monitors. If you change an LCD's native resolution, you will see a badly focused image, and hard-to-read fonts with blurred edges.

The 15-inch and 17-inch models we received had native resolutions of 1024 x 768 and 1280 x 1024 respectively. The 19-inch monitors had a native resolution of 1280 x 1024 when the aspect ratio was 4:3, whereas it was 1440 x 900 when the aspect ratio was widescreen, at 16:9.

The widescreen aspect ratio was only seen in the 19-inch category. The monitors with a wide aspect ratio were the Acer AL1916W, Asus PW191 and the CMV CT-937A. A wide screen is especially good when you're viewing wide-screen movies.

Luminance

Luminance and contrast ratio are very important factors in any monitor, and they are even more

